distributing profits, private equity funds can delay capital calls by first borrowing from fund-level subscription facilities to finance investments.[1211]

This practice has several key implications for investors. First, this practice has been used by private equity funds to artificially boost reported IRRs, but investors must pay the interest on the debt used and can potentially suffer lower total returns.[1212] Second, because the increases to IRR can reflect a manipulation of financing timing (and can distort total returns) rather than being a reflection of the adviser's skill and return opportunities, or even a reflection of the adviser's skill in cash flow management, the higher reported performance can distort fund performance rankings and distort future fundraising outcomes.[1213] Lastly, use of subscription lines to boost IRRs can artificially boost IRRs over the fund's preferred return hurdle rate, resulting in the adviser receiving carried interest compensation in a scenario where the adviser would not have received carried interest without the subscription line, and where the investor may not agree that the subscription line improved total returns and warranted a carried interest payment.[1214] If the use of a subscription line artificially boosts the IRR and does not actually reflect the adviser's investment skill, losses later in the fund's life may be more likely,

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1211 *Id*.

1212 *See, e.g.*, JAMES F. ALBERTUS & MATTHEW DENES, DISTORTING PRIVATE EQUITY PERFORMANCE: THE RISE OF FUND DEBT, FRANK HAWKINS KENAN INSTITUTE OF PRIVATE ENTERPRISE REPORT (June 2019), *available at* https://www.kenaninstitute.unc.edu/wp-content/uploads/2019/07/DistortingPrivateEquityPerformance_07192019.pdf; *Recommendations Regarding Private Asset Fund Subscription Lines*, Cliffwater LLC (July 10, 2017); *Subscription Lines of Credit and Alignment of Interest*, ILPA (June 2017), *available at* https://ilpa.org/wp-content/uploads/2020/06/ILPA-Subscription-Lines-of-Credit-and-Alignment-of-Interests-June-2017.pdf.

1213 *See, e.g.*, Pierre Schillinger, Reiner Braun & Jeroen Cornel, *Distortion or Cash Flow Management? Understanding Credit Facilities in Private Equity Funds* (Aug. 7, 2019), *available at* https://ssrn.com/abstract=3434112; Enhancing Transparency Around Subscription Lines of Credit, *supra* footnote 1001.

1214 Subscription Lines of Credit and Alignment of Interest, *supra* footnote 1212.

potentially resulting in a clawback.[1215] While investors have grown aware of these issues, utilization of subscription lines has continued to grow, and investor industry groups continue to report challenges in achieving visibility into fund liquidity and cash management practices around subscription lines.[1216] As for reinvestment assumptions, the IRR as a performance measure assumes that cash proceeds have been reinvested at the IRR over the entire investment period. For example, if a private equity or other fund determined to be illiquid reports a 50% IRR but has exited an investment and made a distribution to investors early in its life, the IRR assumes that the investors were able to reinvest their distribution again at a 50% annual return for the remainder of the life of the fund.[1217]

Although IRR remains one of the leading standardized methods of reporting returns at present, these and other drawbacks make IRR difficult as a singular return measure, especially for investors who likely may not understand the limitations of the IRR metric, and the differences between IRR and total return metrics used for public equity or registered investment funds.

Several other measures have emerged for measuring the performance of private equity and other funds that would be determined to be illiquid under the final rule. These measures compensate for some of the shortcomings of IRR at the cost of their own drawbacks. Multiple of invested capital ("MOIC"), used by private equity funds, is the sum of the net asset value of the investment plus all the distributions received divided by the total amount paid in. MOIC is

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1215 *Id*.

1216 Enhancing Transparency Around Subscription Lines of Credit, *supra* footnote 1001.

1217 *See, e.g.*, Oliver Gottschalg & Ludovic Phalippou, *The Truth About Private Equity Performance*, HARVARD BUS. REV. (Dec. 2007), *available at* https://hbr.org/2007/12/the-truth-about-private-equity-performance.

simple to understand in that it is the ratio of value received divided by money invested, but has a key drawback that, unlike IRR, MOIC does not take into account the time value of money.[1218]

Another measure closely related to MOIC is the TVPI, or "total value to paid-in capital" ratio.[1219] When applied to an entire fund, MOIC and TVPI are similar performance metrics. However, both MOIC and TVPI have analogous measures than can be applied to just the realized and unrealized portions of a fund, and differ in their approaches to these portions of funds. For TVPI, the unrealized and realized analogues are RVPI ("residual value to paid-in capital") and DPI ("distributions to paid-in capital") ratios, and the denominator in both of these cases is the total called capital of the entire fund.[1220] For MOIC, unrealized and realized MOIC have as denominators just the portions of the called capital attributable to unrealized and realized investments in the portfolio. RVPI and DPI sum to TVPI, while unrealized MOIC and realized MOIC must be combined as a weighted average to yield total MOIC. In the staff's experience, in the TVPI framework, substantial misvaluations applied to unrealized investments, when unrealized investments are a small portion of the fund's portfolio, may go undetected because in that case the denominator in the RVPI will be very large compared to the size of the misevaluation. By comparison, unrealized MOIC will have as a denominator just the called

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1218 One commenter argues that neither IRR nor MOIC takes into account the timing of fund transactions, and provides as an example three funds with different timing of contributions and distributions but the same IRR. *See* XTAL Comment Letter. We disagree. The fact that it is possible to construct examples in which two funds with different timings of payments can have the same IRRs does not mean that IRR broadly fails to take into account the time value of money. Rather, this only indicates that in any such examples, the comparable funds are offering similar performances to their investors, taking the time value of money into consideration. We continue to understand that, in general, IRR takes into account the time value of money.

1219 *See supra* section II.B.2.b).

1220 *See, e.g.*, *Private Capital Performance Terms*, PREQIN, *available at* https://www.preqin.com/academy/industry-definitions/private-capital-performance-terms-definitions.

capital contributed to the unrealized investments, and so the misevaluation may be easier to detect.

Another measure, Public Market Equivalent ("PME"), also used by private equity and other illiquid funds, is sometimes used to compare the performance of a fund with the performance of an index.[1221] The measure is an estimate of the value of fund cash flows relative to the value of a public market index. Relative to a given benchmark, differences in PME can indicate differences in the performance of different private fund investments. However, the computation of the PME for a fund requires the availability of information about fund cash flows including their timing and magnitude.

Regardless of the performance measure applied, another fundamental difficulty in reporting the performance of illiquid funds is accounting for differences in realized and unrealized gains. Illiquid funds generally pursue longer-term investments, and reporting of performance before the fund's exit requires estimating the unrealized value of investments.[1222] There are often multiple methods that may be used for valuing an unrealized illiquid investment. As discussed above, the valuations of these unrealized illiquid investments are typically determined by the adviser and, given the lack of readily available market values, can be challenging. Such methods may rely on unobservable models and other inputs.[1223] Because advisers are typically evaluated (and, in certain cases, compensated) based on the value of these

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1221 *See, e.g.*, Robert Harris, Tim Jenkinson & Steven Kaplan, *Private Equity Performance: What Do We Know?*, 69 J. FIN. 1851 (2014), *available at* https://onlinelibrary.wiley.com/doi/full/10.1111/jofi.12154; Steven Kaplan & Antoinette Schoar, *Private Equity Performance: Returns, Persistence, and Capital Flows*, 60 J. FIN. 1791 (2005), *available* https://onlinelibrary.wiley.com/doi/full/10.1111/j.1540-6261.2005.00780.x.

1222 *See supra* section II.B.2.b).

1223 *Id*.

illiquid investments, unrealized valuations are at risk of being inflated, such that fund performance may be overstated.[1224] Some academic studies have found broadly that private equity performance is overstated, driven in part by inflated accounting of ongoing investments.[1225]

One paper cited by commenters argues that even when advisers do manipulate their investment valuations, "investors can see through [the adviser's] manipulation on average."[1226] Brown et al. (2013) agree that there is evidence of underperforming managers inflating reported returns during times when fundraising takes place, but they also find that, on average, those managers are less likely to raise a subsequent fund.[1227] We disagree with the commenter's assessment that this study indicates that investors in private funds are all equipped to protect their interests without any further regulation. The paper cited itself concedes that in its findings, unskilled investors may misallocate capital, and that it is only the more sophisticated investors who may prefer the status quo to a regime with more regulation.[1228] We believe the commenter's interpretation of this paper also ignores the costs that investors must currently undertake to "see through" manipulation, even on average.

Commenters also added to this discussion that there are different methods and norms for calculating gross performance and then net performance that is net of fees and expenses. In

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1224 *Id*.

1225 *See, e.g.*, Ludovic Phalippou & Oliver Gottschalg, *The Performance of Private Equity Funds*, 22 REV. FIN. STUD. 1747-1776 (Apr. 2009).

1226 AIC Comment Letter I; Gregory W. Brown, Oleg Gredil & Steven N. Kaplan, *Do Private Equity Funds Manipulate Reported Returns?* J. FIN. ECON., Forthcoming, Fama-Miller Working Paper (Apr. 30, 2017) ("Brown et al."), *available at* https://ssrn.com/abstract=2271690.

1227 Brown et al., *supra* footnote 1226.

1228 *Id*.

particular, the CFA Institute described the role of GIPS standards in providing definitions and methods for calculating gross returns and net returns.[1229] The GIPS standards define "gross-of-fees returns" as the return on investments reduced only by trading expenses.[1230] GIPS states that gross-of-fees returns demonstrates the firm's expertise in managing assets without the impact of the firm's or client's skills in negotiating fees.[1231] GIPS defines "net-of-fees returns" as gross-of-fees returns reduced by management fees, including performance-based fees and carried interest.[1232]

The CFA Institute also acknowledged the role of the recent marketing rule in defining gross and net performance.[1233] The marketing rule defines gross performance as "the performance results of a portfolio (or portions of a portfolio that are included in extracted performance, if applicable) before the deduction of all fees and expenses that a client or investor has paid or would have paid in connection with the investment adviser's investment advisory services to the relevant portfolio."[1234] However, the final rule also offers guidance that "the final rule does not prescribe any particular calculation of gross performance . . . Under the final rule, advisers may use the type of returns appropriate for their strategies provided that the usage does not violate the rule's general prohibitions."[1235] Thus, gross reporting under GIPS standards

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1229 CFA Comment Letter I; CFA Comment Letter II.

1230 GIPS, *Guidance Statement on Fees* (Sept. 28, 2010), *available at* http://www.gipsstandards.org/wp-content/uploads/2021/03/fees_gs_2011.pdf.

1231 *Id*.

1232 *Id*.

1233 *See, e.g.*, CFA Comment Letter I; CFA Comment Letter II.

1234 Marketing Release, *supra* footnote 127.

1235 *Id*.

deducts transaction fees, but under the marketing rule may or may not, depending on the adviser's internal calculation methodologies.

The marketing rule defines net performance as "the performance results of a portfolio (or portions of a portfolio that are included in extracted performance, if applicable) after the deduction of all fees and expenses that a client or investor has paid or would have paid in connection with the investment adviser's investment advisory services to the relevant portfolio, including, if applicable, advisory fees, advisory fees paid to underlying investment vehicles, and payments by the investment adviser for which the client or investor reimburses the investment adviser."[1236] Thus, net returns under GIPS standards only deduct management fees, performance-based fees, and carried interest, but under the marketing rule all fees and expenses may be deducted, depending on the adviser's treatment of certain fees and expenses, such as custodian fees for safekeeping funds and securities.

For illiquid funds under the final rules, standard industry methods for reporting performance do not use annual returns, because annual returns for individual years may be substantially less informative for investors. For an investor in an illiquid fund who has limited or no ability to withdraw or redeem from a fund, we understand that the investor's primary concern is more typically measurement of the total increase in the value of its investment over the life of the illiquid fund and the average cumulative return as measured by MOIC and IRR, rather than the annual returns in any given year. Consistent with this, many commenters expressed support

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[1236] *Id.*

for the proposal's rules that would require MOIC and IRR for private equity funds and other illiquid funds, as compared to requiring annual returns.[1237]

Private equity funds and other illiquid funds also must, as discussed above,[1238] more frequently measure performance of the fund both with respect to realized and unrealized investments. In addition to the challenges described above, the difficulty of valuing unrealized investments often contributes to what is deemed a "J-Curve" to illiquid fund performance, causing many performance metrics to report negative returns for investors in early years (as investor capital calls occur, funds deploy capital, and funds hold unrealized investments) and large positive returns in later years (as investments succeed and are exited, and proceeds are distributed).[1239] As discussed above and in the Proposing Release, these problems are exacerbated by a potential lack of reliable valuation data prior to realization of an investment, in particular when the fund primarily invests in illiquid assets.[1240] For investors in those funds, an annual return in the middle of the life of the fund therefore does not provide the same information as the cumulative impact of their investments since the fund's inception, as measured by MOIC and IRR.[1241]

Other approaches tend to be used for evaluating the performance of hedge funds and other liquid funds. In particular, investors who are determining whether and when to withdraw

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1237 *See, e.g.*, OPERS Comment Letter; CFA Comment Letter II.

1238 *See supra* footnote 1222 and accompanying text.

1239 *See, e.g.*, *J Curve*, CORP. FIN. INST. (June 28, 2023), *available at* https://corporatefinanceinstitute.com/resources/economics/j-curve/.

1240 *See supra* footnote 1222 and accompanying text; *see also* Proposing Release, *supra* footnote 3, at 59-60.

1241 Because these problems are exacerbated when the fund primarily invests in illiquid assets, as separate from when the investors' interests in the fund are illiquid, certain funds that will be defined to be liquid funds under the final rules may also rely on IRR and MOIC performance reporting today.

from or request a redemption from a liquid fund typically find annual net total returns more informative than metrics such as an IRR measured since the fund's inception, as annual net total returns allow the investor to measure whether the liquid fund's performance is likely to continue to outperform its next best investment alternative. Consistent with this, many commenters disagreed with the proposed rule requirement of annual net total returns since inception, stating that more recent returns are more relevant.[1242] Other methods include a fund's "alpha" and its "Sharpe ratio." A fund's alpha is its excess return over a benchmark index of comparable risk. A fund's Sharpe ratio is its excess return above the risk-free market rate divided by the investment's standard deviation of returns. Many, but not all, hedge funds disclose these and other performance measures, including net returns of the fund. Many hedge fund-level performance metrics can be calculated by investors directly using data on the fund's historical returns, by either combining with publicly available benchmark index data (in the case of alpha) or by combining with an estimate of the standard deviation of the fund's returns (in the case of the Sharpe ratio). Despite these detailed methods, data in commercial databases on hedge fund performance reporting may also be biased, because hedge funds choose whether and when to make their performance results available to commercial databases.[1243]

1242 *See, e.g.*, ATR Comment Letter; ICM Comment Letter.

1243 *See, e.g.*, Philippe Jorion & Christopher Schwarz, *The Fix Is In: Properly Backing Out Backfill Bias*, 32 SOC'Y FIN. STUD. 5048-5099 (Dec. 2019); *see also* Nickolay Gantchev, *The Costs of Shareholder Activism: Evidence From A Sequential Decision Model*, 107 J. FIN. ECON. 610-631 (2013). One commenter stated that "[t]he Proposed Rule also casts doubt on the reliability of public data on hedge fund performance . . . implying that these data may [] overstate fund performance. The Proposed Rule then suggests that its proposed restrictions will remedy this purported lack of price and quality competition." CCMR Comment Letter IV. We believe this mischaracterizes the Proposing Release. The discussion in the Proposing Release, and in this release, pertain to whether existing private tools are sufficient for investors seeking to evaluate the performance of hedge fund advisers and other liquid fund advisers. The paragraph cited by the commenter discusses that there are limitations to the extent to which investors may be able to conduct complete evaluations of the performance of their adviser using existing methods because, for example, public commercial databases may have biased data. We agree with the commenter that, for example, there is no literature concluding that hedge fund performance is low, and that public data on hedge fund

Because CLOs and other SAFs primarily issue debt to investors, typically structured as notes and issued in different tranches to investors, typical fee, expense, and performance reporting practices for these funds differ from other types of funds.[1244] Typical reporting for SAFs is designed to provide relevant information to different debt tranches of a fund, which offer different defined returns based on different priorities of payments and different defined levels of risk associated with their notes. Because debt interests in a SAF are not structured to provide variable investment returns like an equity interest, SAF reporting metrics prioritize measuring the likelihood of the debt investor receiving its previously agreed-upon defined return. For example, commenters stated that CLOs typically report overcollateralization ratios, examinations of the average credit rating of the portfolio, the diversity of holdings within the portfolio, and the promised yield of portfolio assets.[1245] Monthly reports of the portfolio holdings will also often include one or more credit ratings for each individual asset in the portfolio,[1246] and also often include summaries of cash flows and mark to market valuations for every asset in the portfolio.[1247] Finally, commenters stated that CLO managers typically earn three types of management fees, all of which are set out in the indenture and paid in accordance with the

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performance indicating otherwise are not a reliable rebuttal to assertions of low hedge fund performance. *See* Proposing Release, *supra* footnote 3, at 208, 230. Moreover, the commenter then cites additional literature illustrating that some hedge fund advisers may understate their performance in public commercial databases, for example to prevent disclosing clues about their proprietary trading strategies. We believe this result means the literature demonstrates that there is likely variation in the bias of performance reporting by hedge fund advisers. Variation in the bias of performance reporting by advisers further limits the ability to which commercial databases today can satisfy investor needs when evaluating their advisers, as investors cannot tell the direction of bias of any given adviser in the data.

1244 *See supra* section II.A.

1245 LSTA Comment Letter; SFA Comment Letter I; SFA Comment Letter II; SIFMA-AMG Comment Letter I; TIAA Comment Letter.

1246 *Id*.

1247 *Id*.